Mechanical Engineering Department, CoE, KNUST

ME 161 Basic Mechanics

Unit 4:

Analysis of Structures

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INTRODUCTION

- We examines **analysis of 2 types of engineering structures designed for supporting loads**, namely **trusses** and **frames**
- Trusses and frames are **composed of several interconnected bodies referred to as members**.
- The **members may be metal bars**, **angles and channels or wooden strut**.

- **Truss** is **composed solely of slender members** **connected together by pin-joints** and **designed primarily to support loads applied at joints**

- A **frame may be designed to support both moments and forces applied at any point in the structure**

- The **analysis of trusses and frames is based on the principle that if a structure is in equilibrium then each member of the structure is also in equilibrium**
- **By applying the equations of equilibrium to the various parts of a frame or truss, all the forces acting at the joints can be determined**
- The **analysis can be extended to determine internal forces in members**. However, that is treated under strength of materials

INTRODUCTION

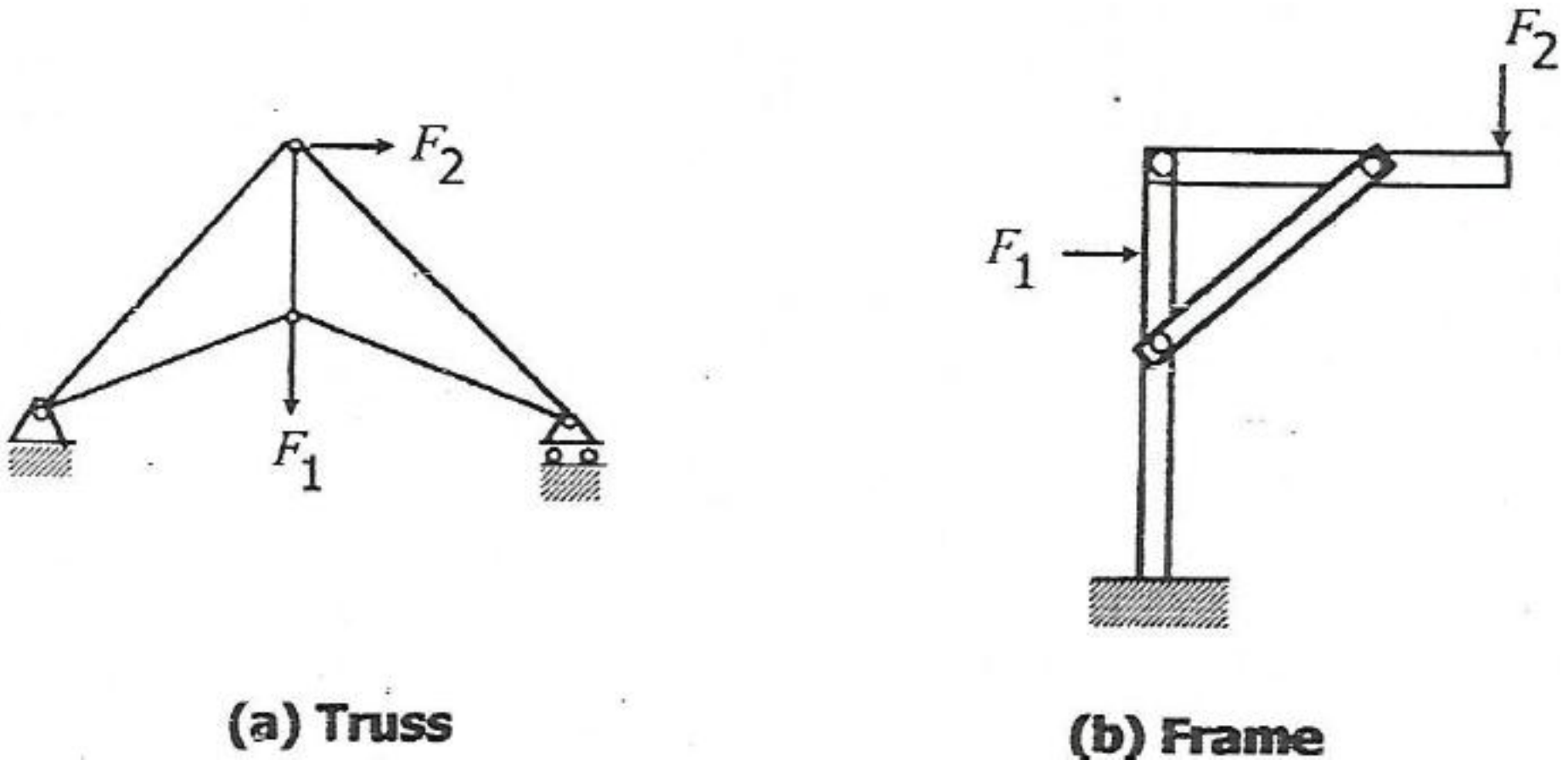


Fig. 3.1 Example of a truss and a frame

Simple Trusses

- In **analyzing trusses it is assumed that the members are connected together by pin joints only**.
- However, in an **actual truss the members may be fastened together by welding, riveting or bolting to a common plate called a gusset plate.**
- It **can be shown that the pin-joint assumption is satisfactory provided that the centre lines of all the members meeting at the connection are concurrent as shown in Fig. 3.2**
- It is also **assumed that the members of the truss are all straight and that all loads are applied at the joints**
- **ALL these assumptions mean that each truss member is a 2–force body**
- Hence, **the resultant of the forces at the 2 ends must be directed along the axis of the member**
- **When these 2 forces tend to elongate the member, the member is said to be in tension and is called a tie (Fig. 3.3a)**
- **When the forces tend to compress the member, the member is said to be in compression and is called a strut (Fig. 3.3b)**
- **Structural designers are very interested in knowing which members are in tension or compression because some materials are strong in compression but very weak in tension**
- Also, **long slender members which can take large tensile loads may buckle under relatively small compressive loads.**

Simple Trusses

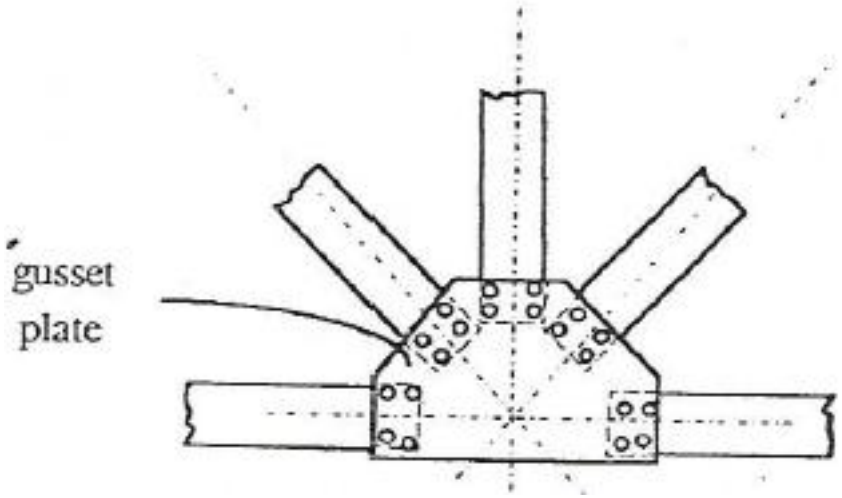


Fig. 3.2 Fastening of truss members by means of a gusset plate

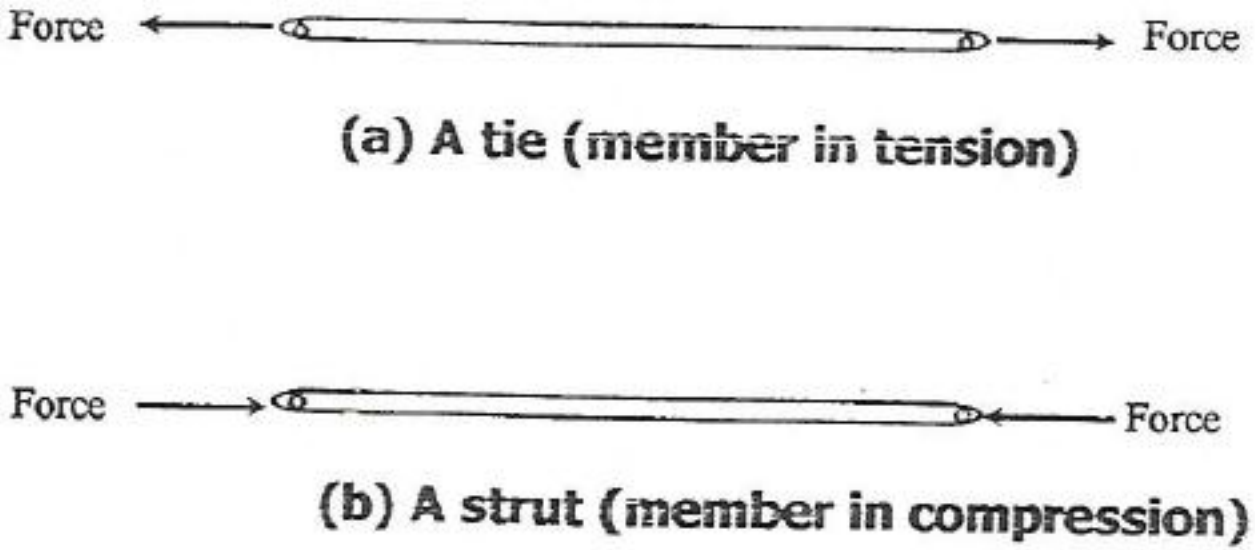


Fig. 3.3 Truss members in tension and compression

- **Very often in the force analysis of trusses the weights of members are neglected since the loads are usually very much larger than the weights**
- However, **where the weight is not negligible a satisfactory analysis can normally be obtained by replacing the weight by 2 vertically downward forces, each equal to half the weight, acting at the ends of the member**
- A **SIMPLE TRUSS** **is constrained by starting with a triangular element, such as ABC in Fig. 3.4. and forming an additional element by successively adding 2 more members at a time, e.g BD and CD in Fig. 3.4**

Simple Trusses

- The **basic triangular element is essential to the construction of rigid trusses**.
- **If an attempt is made to use a polygon having more than 3 sides, the result is a non-rigid structure called a mechanism (Fig. 3.5).**
- In mechanism, the pin joints allow the members to rotate relative to one another

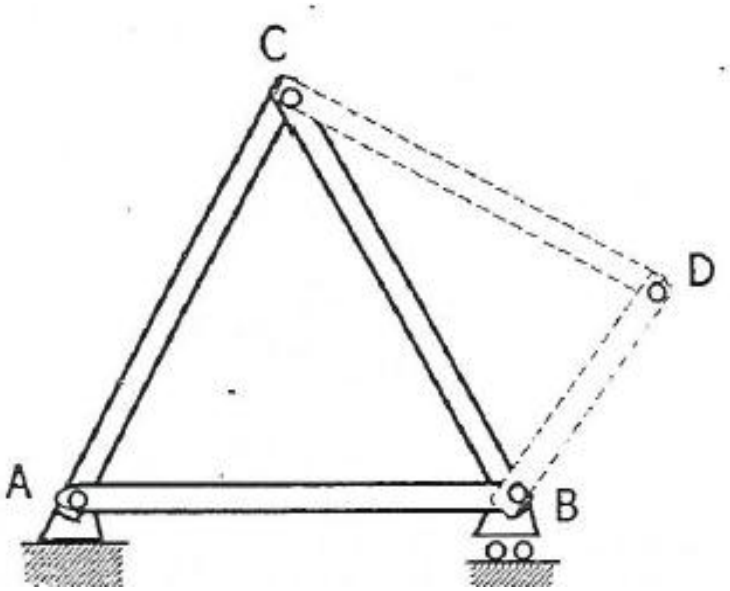


Fig. 3.4 A truss formed by adding to the basic triangular element

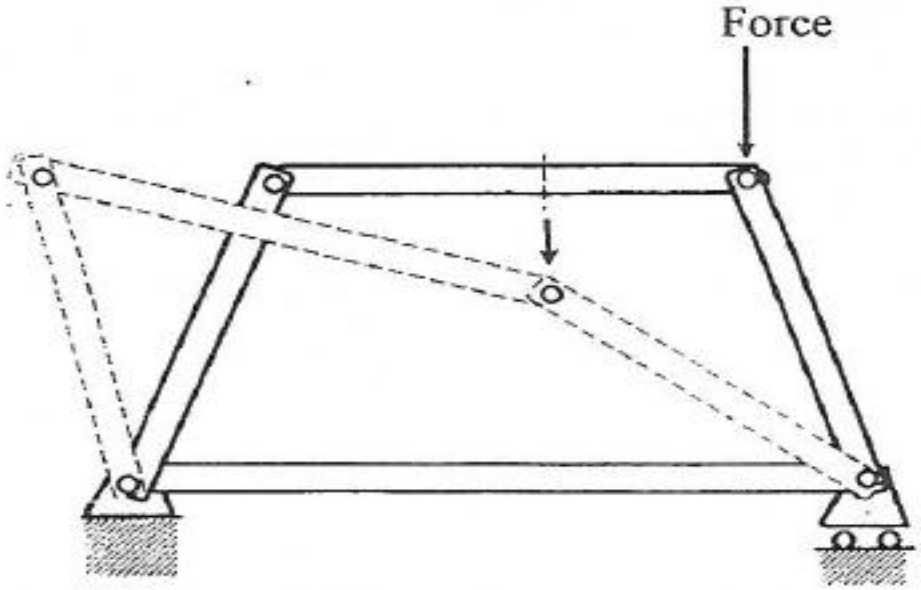


Fig. 3.5 A non-rigid structure (a mechanism)

- **A planar truss is one that lies in a single plane.**
- **Most actual trusses are connected by joining together several trusses to form a space framework.**
- **Each of the trusses is designed to support only those loads which act in its plane and, therefore, the truss may be considered as a planar truss**

Simple Trusses

- The truss shown in Fig. 3.6 is an example of a bridge truss.
- Since the loads must be applied only at the joints and nowhere else, a floor must be provided to transmit all loads to the joints.
- Thus, any load on the deck of the bridge is 1st transmitted to stringers, then to floor beams, and finally to the joints.
- Notice that one end of the truss is supported on rollers.
- This is commonly done when a truss has a long span.
- The roller or rocker allows expansions or contractions that can occur as a result of temperature changes or deformations due to the loading.

Simple Trusses

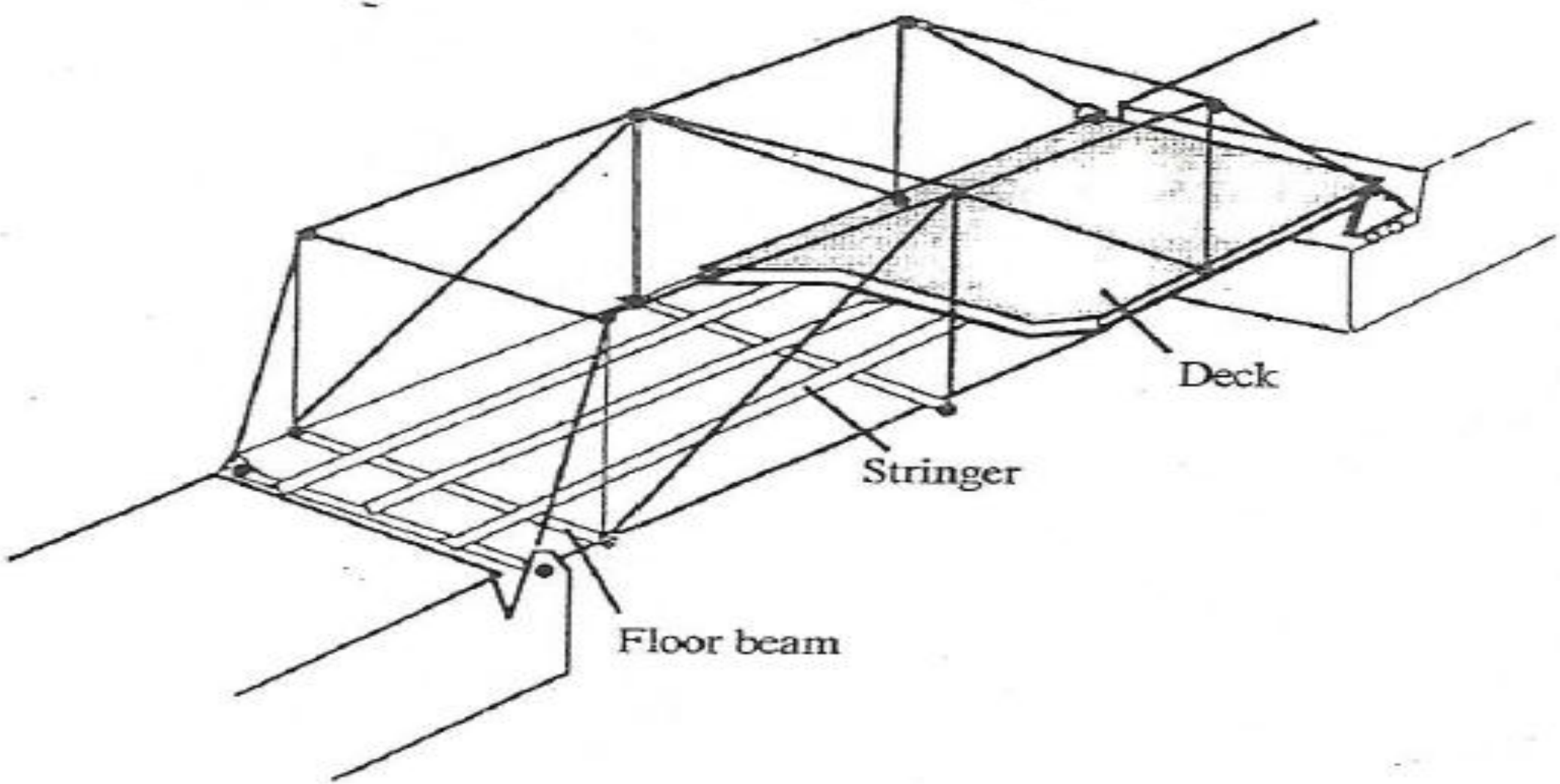


Fig. 3.6 A bridge truss

The Method of Joints

- The **forces in the members of a simple truss may be determined by the method of joints which is based on the fact that if a truss is in equilibrium then so is each of its joints and each of its members.**
- The method consists of **applying the conditions of equilibrium to forces exerted** on the **pins** at each joint.
- **The system of forces acting on each pin is concurrent and coplanar since all truss members are straight 2-force members lying in the same plane.**
- Therefore, **it is only necessary to satisfy the conditions for force equilibrium, (i.e. $\sum F_x = 0$ and $\sum F_y = 0$)**
- In applying the Method of joints **it is necessary to start the analysis at a joint having at least one known force and at most 2 unknown forces**
- **If this joint is at one of the supports it will usually be necessary to know the external reactions.**
- **The reactions at the supports may be obtained by considering the entire truss as a free-body.**
- The **free-body diagram of each pin is then drawn, indicating the forces exerted on the pin by the members or external loads or support to which it may be connected.**
- The **forces exerted by each member on the pins at its ends are equal in magnitude but opposite in sense and are directed along that member.**
- Thus, **only the magnitudes of the forces in the members are unknown.**
- The **magnitudes of the forces can be determined by simply considering the equilibrium of each joint.**

The Method of Joints

- In **drawing the free-body diagrams of the joints it is often possible to determine the correct senses of the unknown forces by inspection.**
- However, **where this is not possible the sense of an unknown force can be assumed, and the correct sense verified from the numerical results obtained from solving the equilibrium equations**.
- A **positive force magnitude indicates that the assumed sense is correct and negative indicates that the assumed sense must be reserved**

Example 4.1

Determine the force in each member of the truss shown in Fig. 4.7 and state whether the members are in tension or compression

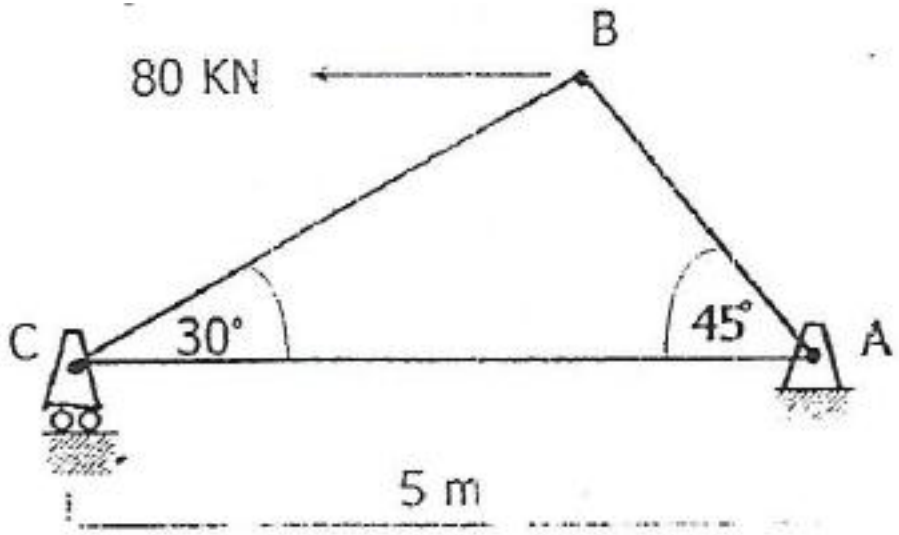


Fig. 4.7 Truss subjected to external loading

SOLUTION

- **We begin the analysis with B since it is the only joint at which there is at least one known force and at most 2 unknown forces.**
- **If we wanted to begin at joint A, for example, it would have been necessary to determine the reaction at that support**

Joint B

The forces acting on the point at B are shown in the free-body diagram in Fig. 4.8(a). The 3 forces acting on the pin are: the external force of 80 kN and the 2 forces exerted by the members BA and BC. Applying the equilibrium equations $\sum Fx = 0$ and $\sum Fy = 0$:

$$\sum F_x = F_{BC}\cos 30^o + F_{BA}\cos 45^o - 80 = 0 \quad \text{or}$$

$$0.8660F_{BC} + 0.7071F_{BA} = 80 \qquad (i)$$

$$\sum F_y = F_{BC}\cos 60^o - F_{BA}\cos 45^o = 0 \quad \text{or}$$

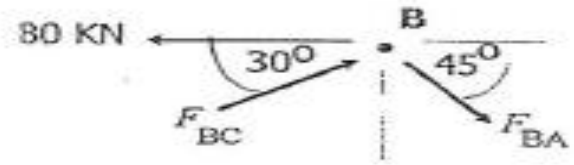
$$0.5F_{BC} - 0.7071F_{BA} = 0 \qquad (ii)$$

Solving equations (i) and (ii) simultaneously

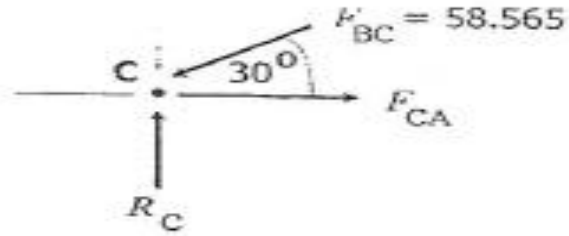
$$F_{BC} = \frac{80}{1.366} = 58.565 \text{ kN (compression)} \quad \text{and}$$

$$F_{BA} = \frac{0.5}{0.7071}(58.565) = 41.412 \text{ kN}$$

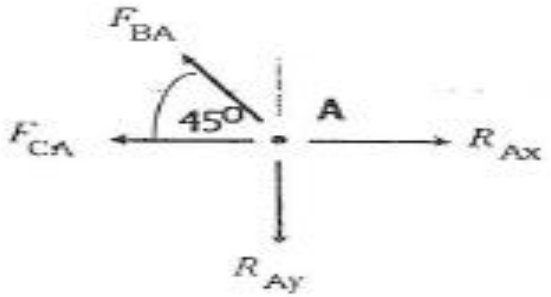
With the forces in members BC and BA determined, we can now use the results to analyse the forces at joints A and C



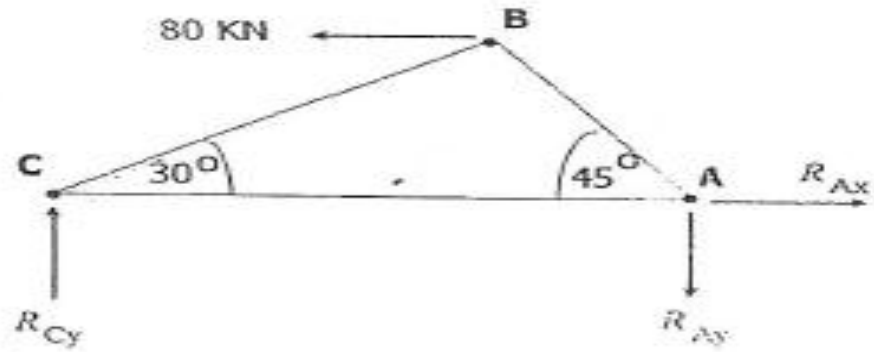
(a) Free-body diagram of pin at B



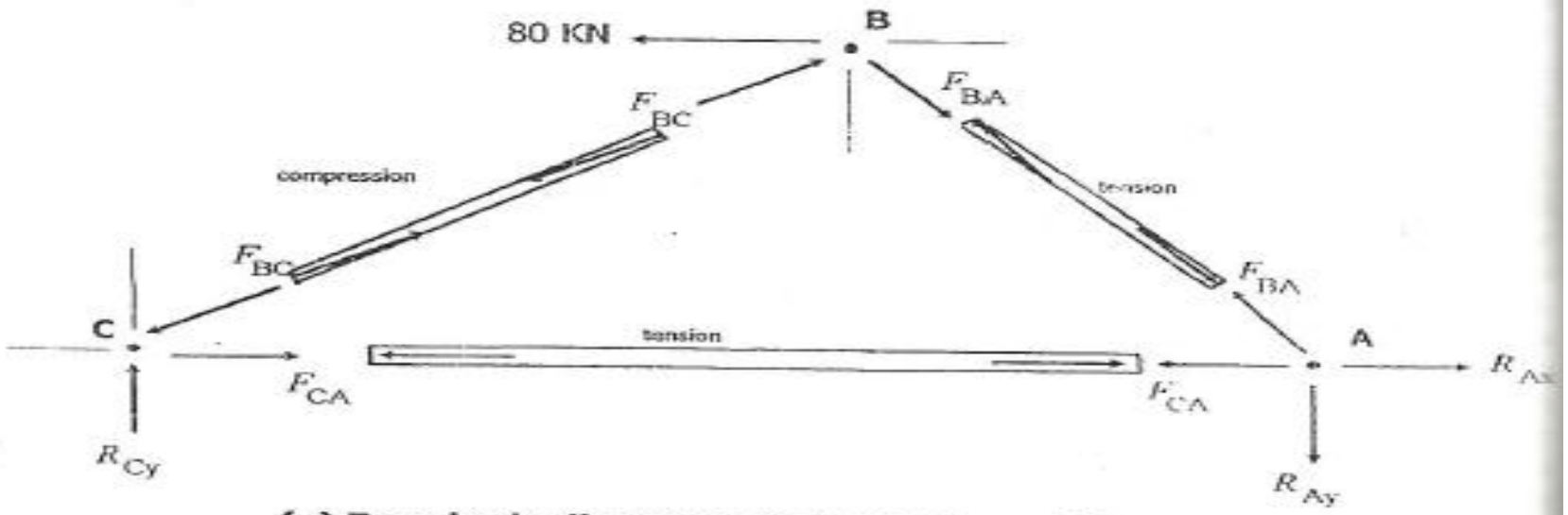
(b) Free-body diagram of pin at C



(c) Free-body diagram of pin nt A



(d) Free-body of whole truss



(e) Free-body diagrams of all members of truss

Fig. 4.8 Truss subjected to external loading

Joint C

- The free-body diagram for the point at is shown in Fig. 4.8(b). The **2 unknown forces are F_{CA} in member CA and the reaction R_C at the support**
- Applying the equilibrium equations,

$$\sum F_x = 0: \qquad F_{CA} = 58.565\cos 30^o = 50.719\,\text{kN}$$

$$\sum F_y = 0: \qquad R_C = 58.565\cos 60^o = 29.283\,\text{kN}$$

Joint A

The **analysis of joint A is unnecessary since the forces in all the members have been determined**. **Nevertheless, we shall analyse joint A as a check.** The free-body diagram is shown in Fig. 4.8 (c). The x and y components of the reactions at A are R_{Ax} and R_{Ay}, respectively. Applying the equilibrium equations,

$$R_{Ax} = 50.719 + 41.412\cos 45^o = 80\,\text{kN}$$

$$R_{Ay} = 41.412\cos 45^o = 29.283\,\text{kN}$$

We can check whether these reactions are correct by considering the equilibrium of the whole truss, the free-body diagram of which is shown in Fig. 4.8(d). Applying the equilibrium equations to the whole truss,

$$\sum F_x = 0 \qquad R_{Ax} = 80\ kN$$

$$\sum F_y = 0 \qquad R_{Ay} = R_C = 29.283\ kN$$

The results of the analysis are shown in Fig. 4.8(e). The forces acting on each pin and on each member are shown in Fig. 4.8. **Members AB and AC are in tension** while **member BC is in compression**. Notice that a member in tension "pulls" on the pins at its ends, and a member in compression "pushes" on the pins

Zero-force Members

- The **analysis of trusses by the method of joints is often simplified** if one first identifies **the zero force members** (i.e, **members which support no load at all**). The following general rules may be used:

(a) If **a truss joint is formed by only 2 members**, **and the joint is not subjected to any external load or support reactions, then the 2-members must be zero-force members**

(b) If a **truss joint is formed by 3 members 2 of which are collinear**, **the 3rd member will be a zero-force member provided that no external load or support reaction is applied at the joint**

- These general rules may be deduced from considerations of force equilibrium of the joints under the special loading conditions specified

Zero-force Members

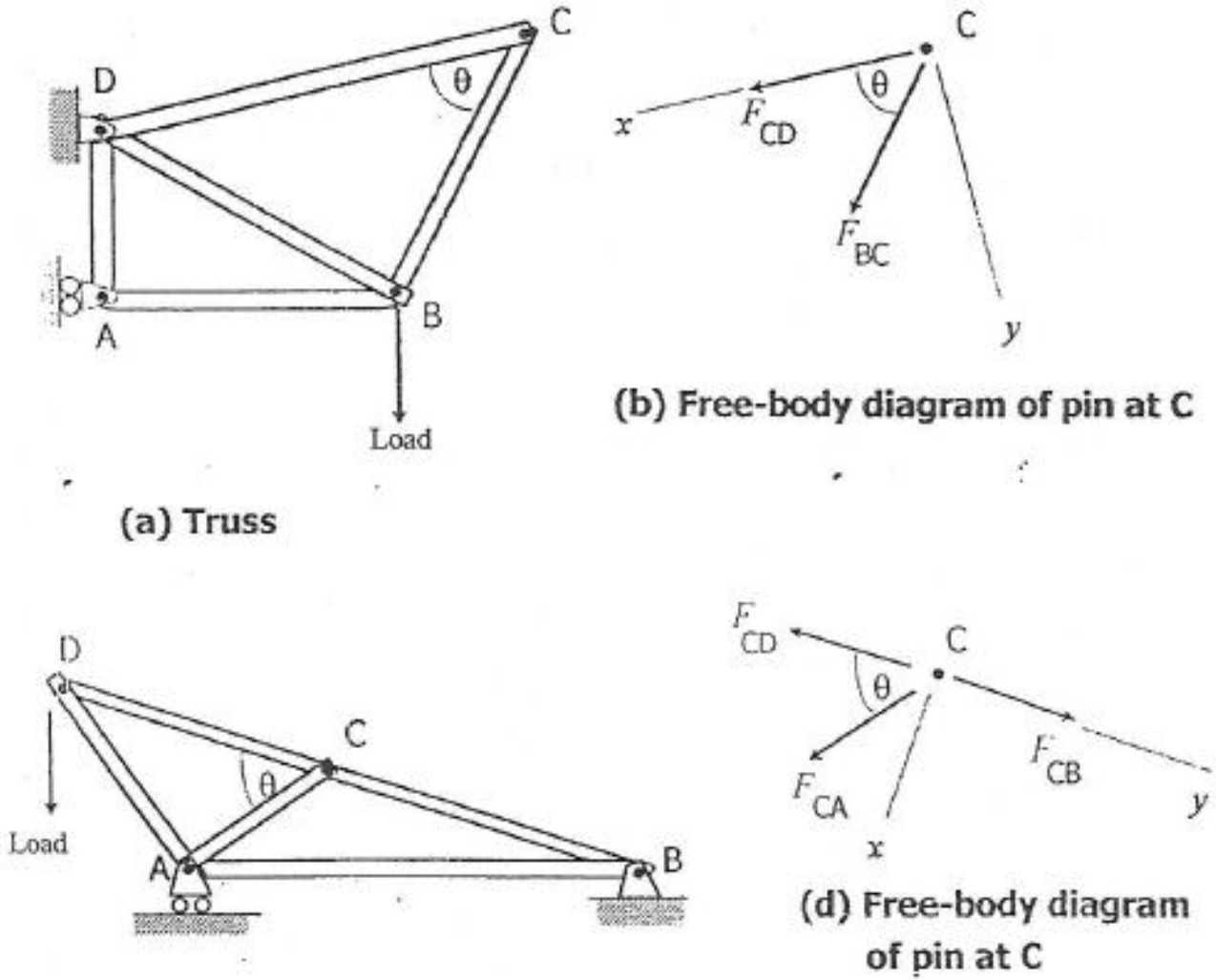


Fig. 4.9

- In the truss in Fig. 4.9 (a), for example, if we consider the free-body diagram of the **pin at joint C** (Fig. 4.9c), it is obvious that members BC and CD are zero-force members since

$$\sum F_y = 0 \Rightarrow \quad F_{BC}\sin\theta = 0 \quad \Rightarrow F_{BC} = 0 \quad \text{since} \quad \sin\theta \neq 0$$

$$\sum F_x = 0 \Rightarrow \quad F_{CD} + F_{BC}\cos\theta = 0 \quad \Rightarrow F_{CD} = 0 \quad \text{since} \quad F_{BC} = 0$$

In the truss shown in Fig. 4.9c the **member AC is a zero-force member** because from the free-body diagram, Fig. 4.9d, we have

$$\sum F_x = 0 \Rightarrow \quad F_{CA}\sin\theta = 0 \quad \Rightarrow F_{CA} = 0 \quad \text{since} \quad \sin\theta \neq 0$$

$$\sum F_y = 0 \Rightarrow \quad F_{CB} - F_{CD} - F_{CA}\cos\theta = 0 \quad \Rightarrow F_{CB} = F_{CD} \quad \text{since} \quad F_{CA} = 0$$

Notice that the forces in members CB and CD are equal

Zero-force Members

- **Another result which may be used to expedite the analysis of trusses is that: if an unloaded joint connects 4 members lying in 2 intersecting lines (see Fig. 4.10) the forces in opposite members must be equal**. This is obvious since

$$\sum F_y = 0 \Rightarrow \quad F_{AC}\sin\theta - F_{AE}\sin\theta = 0 \quad \Rightarrow F_{AC} = F_{AE}$$

$$\sum F_x = 0 \Rightarrow \quad F_{AB} + F_{AC}\cos\theta - F_{AE}\cos\theta - F_{AD} = 0 \quad \Rightarrow F_{AB} = F_{AD}$$

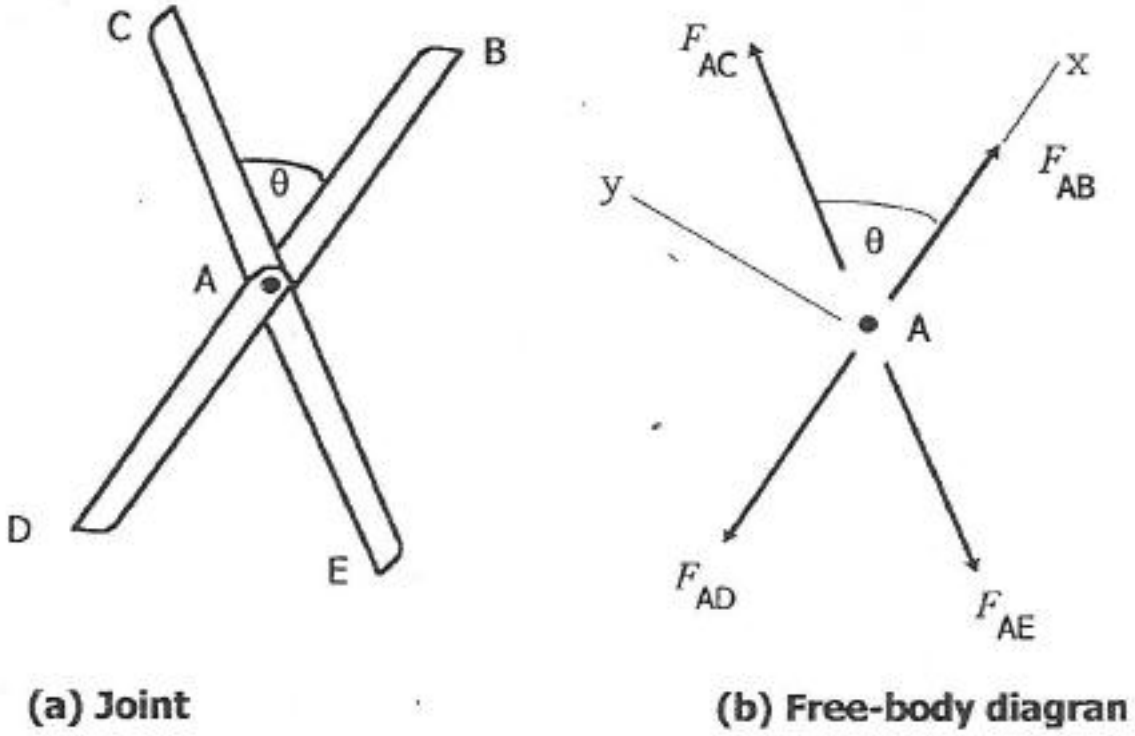


Fig. 4.10 Joint with 4 members lying in 2 intersecting lines

NOTE:

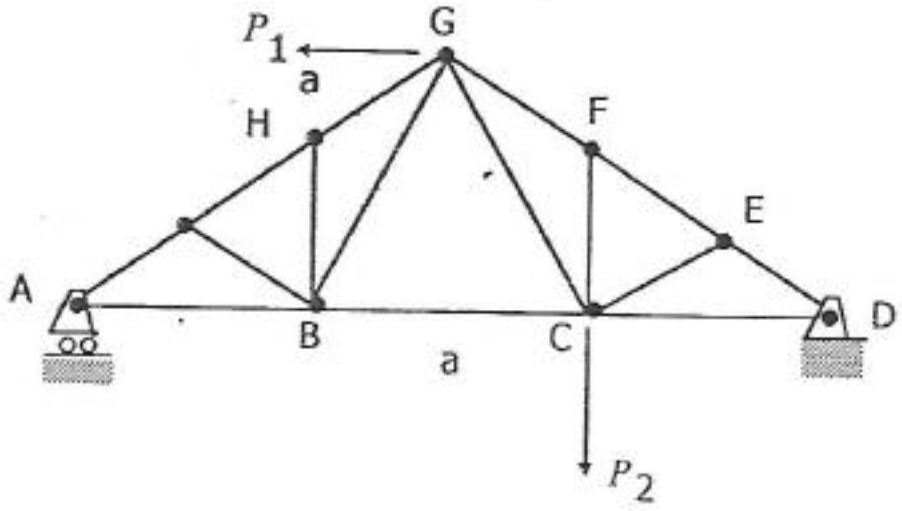
- **Zero-force are not useless.**
- They may be **needed to keep the truss stable during construction.**
- **May be required in case the loading on the truss is changed.**
- **Zero-force members are also needed to support the weight of the truss and to keep the truss in the shape desired**

The Method of Sections

- If **the force in only one member or the forces in a few members of a truss are to be determined, the use of the method of joints can become inconvenient**.
- Method of sections comes handy: it is **based on the principle that if a body is in equilibrium, then any part of the body is also in equilibrium.**
- To apply the method of sections, **a section must 1st be passed through the member whose force is to be determined.**
- The **free-body diagram for one of the 2 parts of the truss is then drawn**.
- The **free-body diagram must show the forces exerted by all the members that have been cut by the section.**
- Finally, **the equations of equilibrium are applied to the part in order to determine the forces in the members at the section**.
- Consider, for example, the truss shown in Fig. 4.11(a) and **suppose that it is required to find the force in member HG.**
- We pass a section **a-a** through the truss as shown **so that it cuts the members at HG, BG and BC (Usually, a section must intersect no more than 3 members).**

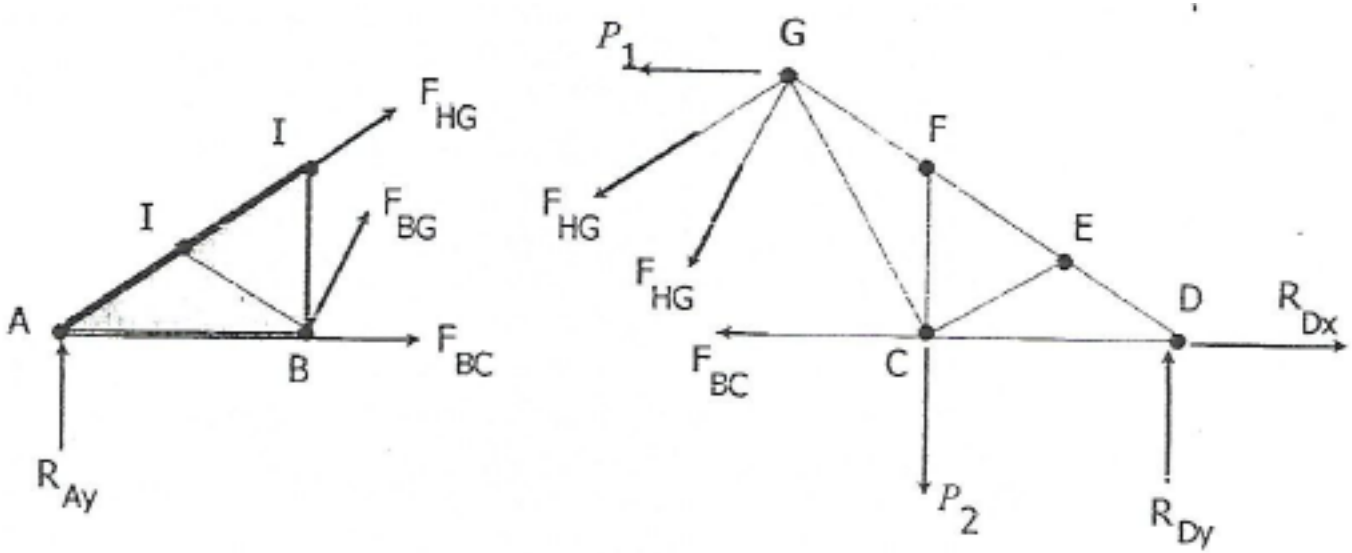
The Method of Sections

- The **free-body diagrams of the 2 parts of the truss are shown in Fig. 4.11b**. We **could use either of the 2 parts to solve the problem.** Here, we shall **consider the left part.**
- **In order to determine F_{HG} we can perform $\sum M_B = 0$. If we were interested in F_{BG} we could perform $\sum M_A = 0$.**
- **If we wanted to determine F_{BC} we could apply the condition $\sum F_x = 0$ after finding F_{HG} and F_{BG}.** As with the method of joints, the correct sense of unknown forces can often be found by inspection.
- However, where this is not possible the sense of unknown forces can be assumed when drawing the free-body diagrams.



(a) Truss showing section a-a

Fig. 4.11 Use of method of sections



(b) Free-body diagrams of parts

Fig. 4.11b Use of method of sections

Example 4.2

Determine the force in **members IB and BC** of the truss shown in Fig. 4.12

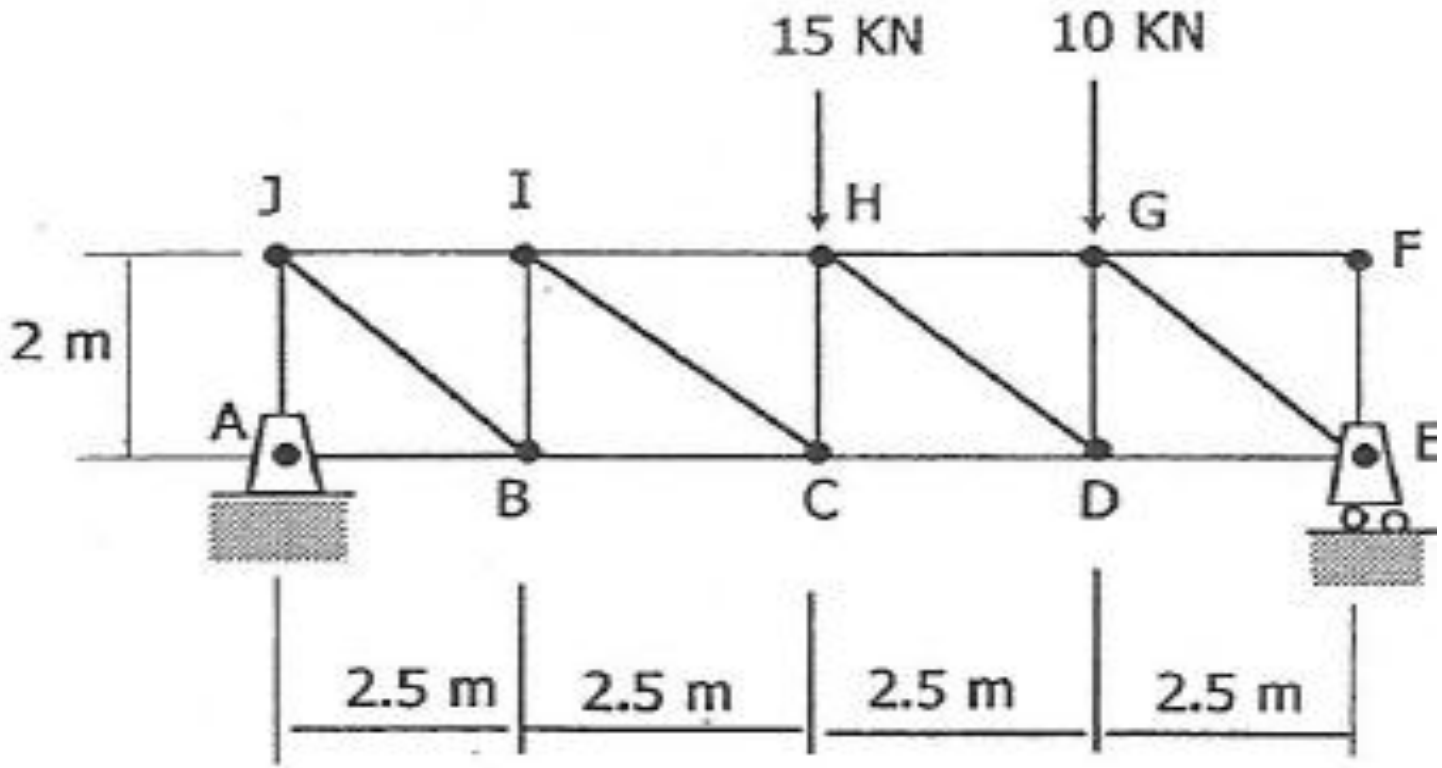
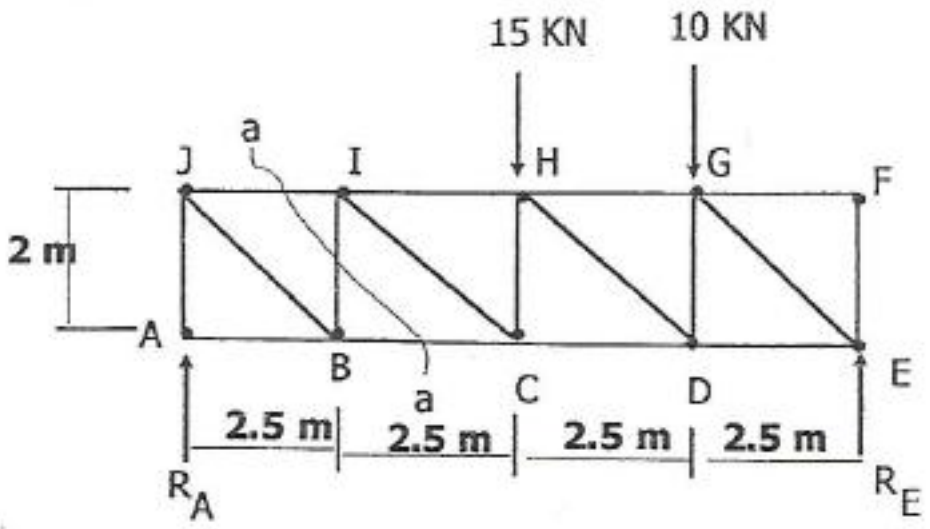
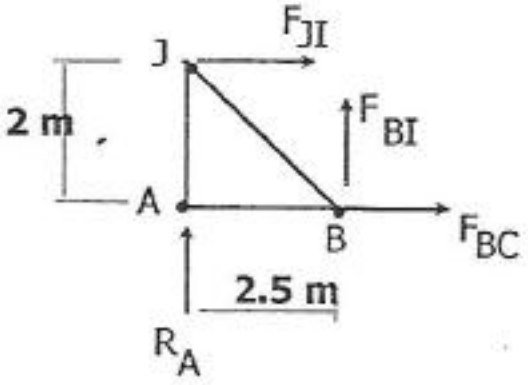


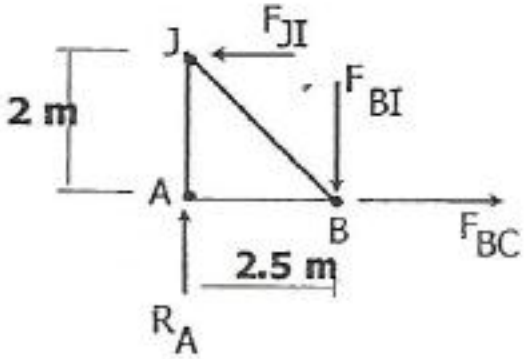
Fig. 4.12 Truss for Example 4.2



(a) Free-body diagram of whole truss



(b) Free-body diagram of left part with assumed directions of forces



(c) Free-body diagram with correct directions of forces

Fig. 4.13 Free-body diagram for truss in Example 4.2

- Pass a section through the truss as shown in Fig. 4.13a. **The reaction R_A at A must be determined separately by considering the free-body diagram of the whole structure as shown in Fig. 4.13b. Taking moments about E, we have**

$$R_A \times 10 = (15 \times 5) + (10 \times 2.5) \quad \Rightarrow \quad R_A = 10\ kN$$

Considering the left part of the truss as shown in Fig. 4.13b, and taking moments about B,

$$R_A \times 2.5 + F_{JI} \times 2 = 0 \quad \Rightarrow \quad R_{JI} = \frac{-R_A \times 2.5}{2.0} = \frac{-10 \times 2.5}{2.0} = -12.5\,kN$$

The **negative sign shows that the correct direction of the force F_{JI} is supposed to be opposite to that shown in Fig. 4.13 (member JI is in compression)**

Summing horizontal forces: $F_{BC} = -F_{JI} = 125$ kN

Finally, summing vertical forces: $F_{BI} + R_A = 0$. Thus, $F_{BI} = -R_A = -10$ kN

-ve sign shows the right direction of F_{BI} is opposite that shown. Therefore, BI is in compression

FRAMES

- A **frame is distinguished from a truss only by the fact that *loads are not necessarily applied at the joints.***
- The **members of a frame are not necessarily 2-force members, and some of the members may be subjected to transverse loads and bending.**
- The **general procedure for the force analysis of frames is to consider simultaneously the equilibrium of the whole frame and each of its members individually.**
- **In drawing the free-body diagram, care must be taken to ensure that Newton's 3rd law is satisfied (i.e, the force exerted by one member on a second member must be equal and opposite to that exerted by the 2nd member on the 1st)**

EXAMPLE 4.3

Determine the force in the pin at C for the pinned frame shown in Fig. 4.14. The weights of the members are negligible

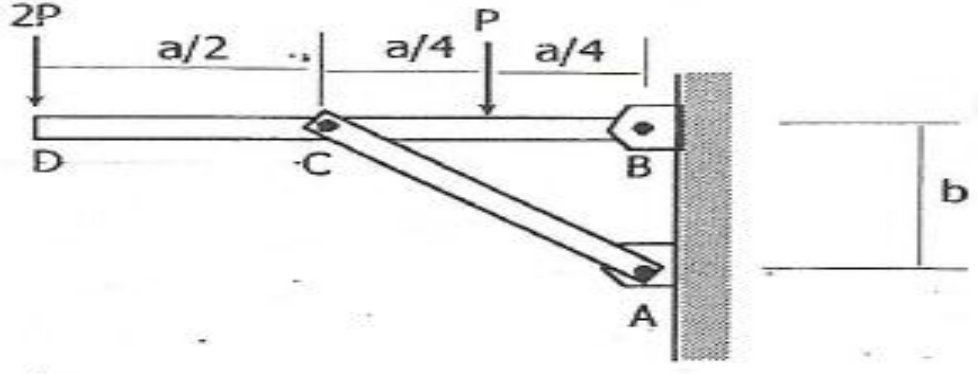
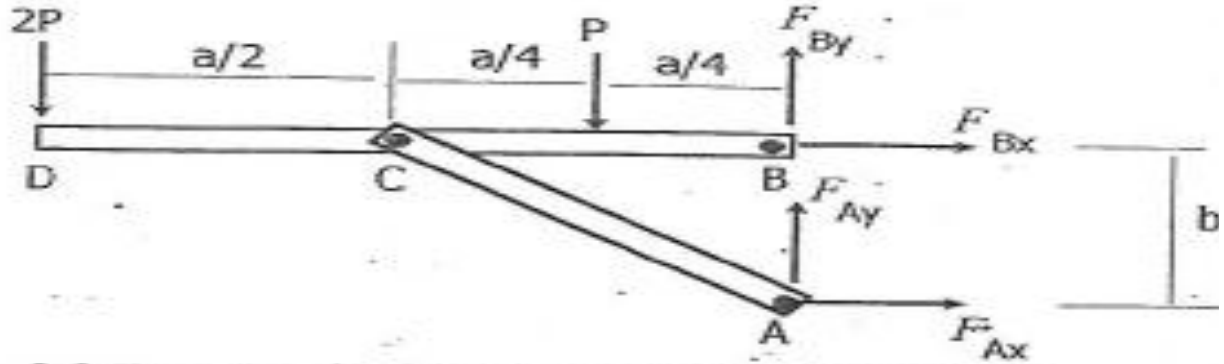
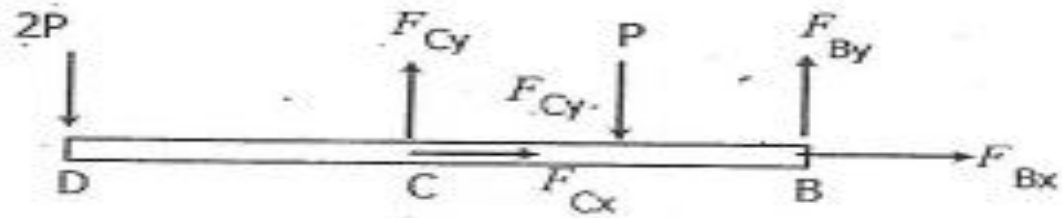


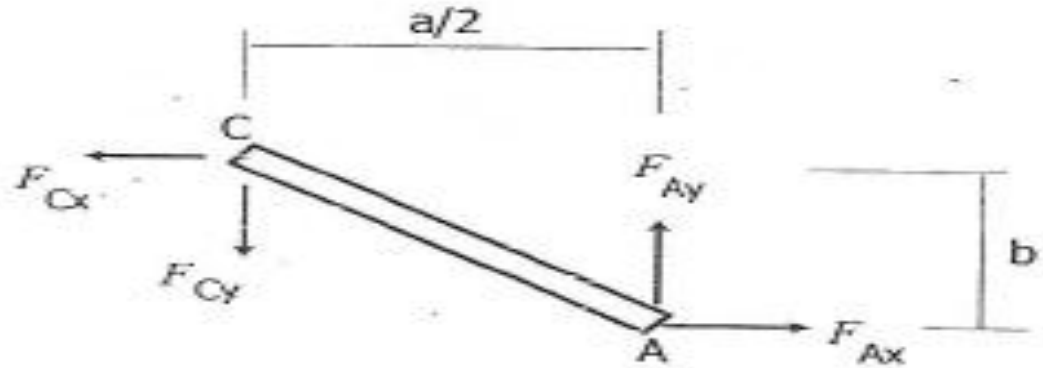
Fig. 4.14 Frame for Example 4.3



(a) Free-body diagram of complete frame



(b) Free-body diagram of member BCD



(c) Free-body diagram of member AC

Fig. 4.15 Free-body diagrams for Example 4.3

SOLUTION

Considering the free-body diagram of the complete frame (Fig. 4.15a), and **taking moments about A:**

$$P \times a/4 + 2P \times a = F_{Bx} \times b \qquad \Rightarrow \qquad F_{Bx} = \frac{9Pa}{4b}$$

Now, **taking the free-body diagram of member BCD (Fig. 4.15b), and summing forces in the horizontal direction:**

$$F_{Cx} = F_{Bx} = \frac{9Pa}{4b}$$

Taking moments about point C for member BCD:

$$2P \times a/2 = P \times a/4 + F_{By} \times a/2 \qquad \Rightarrow \qquad F_{By} = \frac{3P}{2}$$

Finally, **considering the vertical equilibrium of member BCD**:

$$F_{Cy} = 2P + P + F_{By} = 2P + P + \frac{3P}{2} = \frac{9P}{2}$$

- Notice that **it has not been necessary to use the free-body diagram of member AC because it was not required to determine the force at A**.
- However, the free-body diagram of AC could be used to check the results
- As with the analysis of trusses, **it is important to ensure that the equilibrium equations take account of the correct senses of unknown forces whose senses have been assumed in the free-body diagrams.**
- A good way of doing this is to write the equilibrium equations based on the assumed senses, and then carry out all computations maintaining the signs of the numerical values of the forces.
- **At the end of the computations a negative value for a force indicates that the correct sense of the force is opposite to that assumed in the free-body diagram**

Thank You